

1 Introduction

"Today prospectors must deploy complex industrial methods to extract microscopic bullion particles out of tons of earth. That does not mean that gold production is at historical lows" by Marcos López de Prado sparked my interest to dive more into financial mathematics. Pairs trading exploits temporary mispricing between two statistically cointegrated securities. When the z-score of the cointegrated spread exceeds a predefined threshold, the strategy simultaneously purchases the undervalued asset and sell short the overvalued asset in proportion to the estimated hedge ratio, expecting both to converge back to equilibrium.

This project implements and evaluates the pairs trading strategy on S&P 500 constituents, using correlation and cointegration-based pair selection, beta-weighted position sizing, and Bayesian hyperparameter optimization. Utilizing the optimized parameters, the strategy is evaluated over an out of sample test period to assess its real-world performance.

2.1 Data

The security universe is sourced from the S&P 500 index, mainly considering that it is a well-studied list, and its sectors create natural pairs. The strategy firstly scrapes the Wikipedia page of the S&P 500 list and extracts the constituent tickers. Then, it uses the yfinance library to download the stock data of the test and the train periods.

The model is evaluated in three periods:

- Train: 2012~2014, Test: 2015~2018
- Train: 2015~2017, Test: 2018~2021
- Train: 2018~2021, Test: 2021~2024

These three periods were dedicated to assess the generalizability of the strategy across different market regimes, with each window kept to three years to prioritize recent market conditions over longer historical data. It is worth mentioning that it is not necessary to train the model over a long period as demonstrated above. Within each test period, pair selection and hedge ratios are re-estimated monthly, allowing the model to adapt to changing market conditions while keeping the optimized hyperparameters fixed. A long test period would also let stale market regimes dominate the evaluation. The training window length was fixed rather than optimized to avoid overfitting the hyperparameter search itself.

2.2 Methodology

2.2.1 Correlation Filter

The most fundamental statistical indicator measuring the linear relationship of two variables would be the correlation. For random variables X and Y , their correlation is defined as:

$$\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

Where σ_X and σ_Y are standard deviations of X and Y , and $\text{cov}(X,Y)$ is the covariance of X and Y .

The stock prices are firstly transformed into log returns. This is because price levels follow a random walk, which is non-stationary, but log returns are approximately stationary. Log returns are computed as follows:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$$

The pairwise correlation is then constructed across all S&P 500 constituents. Pairs whose correlation exceeds a predefined threshold are retained for further analysis.

2.2.2 Cointegration Test

To check whether two variables are cointegrated using Augmented Dickey-Fuller test (ADF test), we must first compute the residuals of the two variables. By performing an Ordinary Least Squares regression on random variables X and Y , we can obtain its residuals Z_t :

$$\hat{Z}_t = Y_t - \hat{\alpha} - \hat{\beta}X_t$$

Then, the stationarity of the residuals is tested. The ADF test evaluates the presence of unit root in the residuals, indicating the non-stationarity. If a unit root is present, we fail to reject the null hypothesis, implying the residuals are non-stationary and therefore that the two series are not cointegrated. Therefore, we define the null hypothesis as:

H_0 : The series are not cointegrated.

H_1 : The series are cointegrated.

If the p-value of the ADF test is below a predefined threshold, we reject the null hypothesis and conclude that the series are cointegrated.

Given a correlated pair, the strategy performs regression in both directions, Y on X and X on Y , and keeps a lower p-value to catch cointegration regardless of direction, though this introduces a mild bias toward selecting pairs: keeping the better of two tests makes the effective

threshold looser than the nominal value, so some pairs pass that a single fixed-direction test would have rejected. Note that the strategy restricts $|\hat{\beta}| > 0.1$ for stable hedge ratios.

2.2.3 Z-Score Computation

Given that the residuals of the cointegrated pairs are defined as follows:

$$\hat{Z}_t = Y_t - (\hat{\alpha} + \hat{\beta}X_t)$$

We can use the z-score to standardize the residuals as follows:

$$z_t = \frac{\hat{Z}_t - \mu_t}{\sigma_t}$$

Where μ_t and σ_t denote the sample mean and the sample standard deviation of the residuals. If we observe $|z_t|$ is above a predefined threshold, then we simultaneously purchase the undervalued asset and sell short the overvalued asset in proportion to the estimated hedge ratio.

The strategy utilizes a 30-day rolling window of residuals, mean, and standard deviation to compute the z-score. The rolling window allows the equilibrium of the spread to drift over time, but a shorter window causes the z-score to be more vulnerable to noise, while a larger window makes the z-score slower to react, delaying entry and exit signals.

One problem arising from using a 30-day rolling window is that the z-scores cannot be computed for the first 30 days. To compute valid z-scores from the first test day, the 30 training days preceding the test period are used solely to initialize the rolling window and are then excluded from the signal set, avoiding any look-ahead bias.

2.2.4 Beta-Weighted Position Size

If there exists a pair whose rolling z-score exceeds a predefined threshold, the strategy must compute the position in proportion to the estimated hedge ratio. The two legs are held in share ratio β , normalized so the combined position fits the per-trade capital budget. Considering the residual definition from 2.2.2, given Y_t is undervalued and X_t is overvalued (the position reverses when the spread is stretched the other way), for every Y_t we buy, we can hedge the movement by β amount of X_t short.

The strategy also considers z-score in determining the position size by multiplying the following scale to positions sizes for both X and Y :

$$\text{scale} = \min\left(\left|\frac{\hat{Z}_t - \mu_t}{2\sigma_t}\right|, 2\right) = \min\left(\frac{|z_t|}{2}, 2\right)$$

Where z_t denotes the z-score between X and Y . The minimum was considered to restrict how much a single weighted position can grow.

Note that the strategy does not consider duplicated positions. To elaborate, given ticker X and ticker Y are being held, if there exists a new signal between two cointegrated pairs X and Z , the strategy ignores the new signal.

2.2.5 Exit Conditions

The strategy has four distinct exit types, each triggered by a different condition.

1. Exit – normal / profit taking

The spread has reverted, and the residuals are converging to equilibrium. The position closes once the z-score falls back within the predefined exit threshold (i.e., $|z_t| \leq \text{exit_condition}$). This is the intended, “the trade worked” exit.

2. Burst – stop-loss / divergence

The spread diverged further instead of reverting. The position closes once the z-score of the residuals exceeds the predefined burst condition. This is a risk control against a pair that is diverging instead of mean-reverting, essential to limit losses. Note that the Optuna search forces burst into the range $\text{entry_condition} + [0.5, 1.5]$, such that the stop is structurally wider than the entry.

3. Time Limit

The position is held longer than the predefined parameter, *max_holding_days*, regardless of z-score. In relation to the triple-barrier method of López de Prado, the time limit acts as a vertical barrier: if a position neither reverts (exit) nor diverges sharply (burst) within the window, it is closed anyway. This guards against pairs that have quietly stopped mean-reverting without triggering a clear signal, and prevents capital from being tied up indefinitely.

4. Cointegration Lost

At monthly recompute, if a position currently being held contains pairs that are no longer considered to be cointegrated, the position is force-closed. A pair that was cointegrated last month may not be cointegrated this month, so continuing to trade it would rely on a relationship that no longer holds.

The first three exits correspond to the triple-barrier method; the fourth is specific to the rolling re-estimation.

2.2.6 Rolling Monthly Recomputation

As briefly mentioned in the previous section, on a monthly basis, the strategy recomputes the cointegration relationship for all S&P constituents, adding and removing the cointegrated pairs. The recomputation is done monthly specifically so that it is frequent enough to catch regime changes and drop broken pairs, yet infrequent enough that the strategy does not constantly churn the cointegrated pair list.

Each recomputation uses a trailing 252-day (or one trading year) window, which is enough data for a stable Augmented Dickey-Fuller test, yet recent enough that the relationships reflect the current regime. If a pair is determined to be no longer cointegrated, no further action is needed when no position in that pair is open; if a position is held, it is force-closed as described in the 'Cointegration Lost' section above.

Hence, the strategy goes through the following loop: it computes the correlation matrix, filters the pairs with correlation above the predefined threshold, performs a cointegration test, which produces the hedge ratios, then computes the z-score from them. Note that only the pair selection and hedge ratios are updated; the Optuna-optimized hyperparameters remain fixed across the entire test period.

2.3 Parameter Optimization

There were a total of seven parameters to be considered, and the values of these parameters would control the overall performance of the strategy. The strategy optimizes these parameters on the whole training period, then holds them constant throughout the test period.

Optuna uses a tree-structured Parzen estimator, building a probabilistic model of which parameter regions produce good objectives and samples from the regions preferentially. With seven parameters, grid search and random search would be infeasible across a seven-dimensional space. The Optuna search was guided by the following conditions:

Parameter	Conditions	Description
Correlation filter threshold	[0.75, 0.95]	Minimum log-return correlation to retain a pair.
p-value threshold	[0.01, 0.05]	ADF p-value below which a pair is deemed cointegrated.
Entry condition	[1.5, 3.5]	Applied to z-scores.
Exit condition	[0.1, Entry condition – 0.5]	Applied to z-scores.
Burst condition	[Entry condition + 0.5, Entry condition + 1.5]	Applied to z-scores.
Maximum position holding days	[15, 60]	Maximum holding days of positions.
Position size	[0.01, 0.2]	Fraction of total capital allocated per trade.

In between these parameter spaces, the objective of the Optuna model is to maximize a simplified trade-level Sharpe ratio after slippage. The daily series is built from realized P&L on trade-exit dates (zero-filled otherwise) rather than a mark-to-market equity curve, so it approximates rather than exactly measures daily volatility.

$$\text{Sharpe} = \sqrt{252} \frac{\text{mean}(\text{daily net PnL})}{\text{std}(\text{daily net PnL})}$$

Maximizing Sharpe ratio rewards risk-adjusted consistency, while penalizing strategies that rely on a few volatile, lucky trades. If a trial produces fewer than 10 trades or zero variance, its Sharpe is set to $-\infty$, preventing the optimizer from selecting a degenerate parameter set that scores well on a handful of lucky trades.

The train/test split prevents evaluating on training data, and the 500-trial optimization still overfits the training window to some extent, but the strategy defends against this by only reporting out-of-sample test results and by checking consistency across three distinct market regimes, as shown in Section 3.

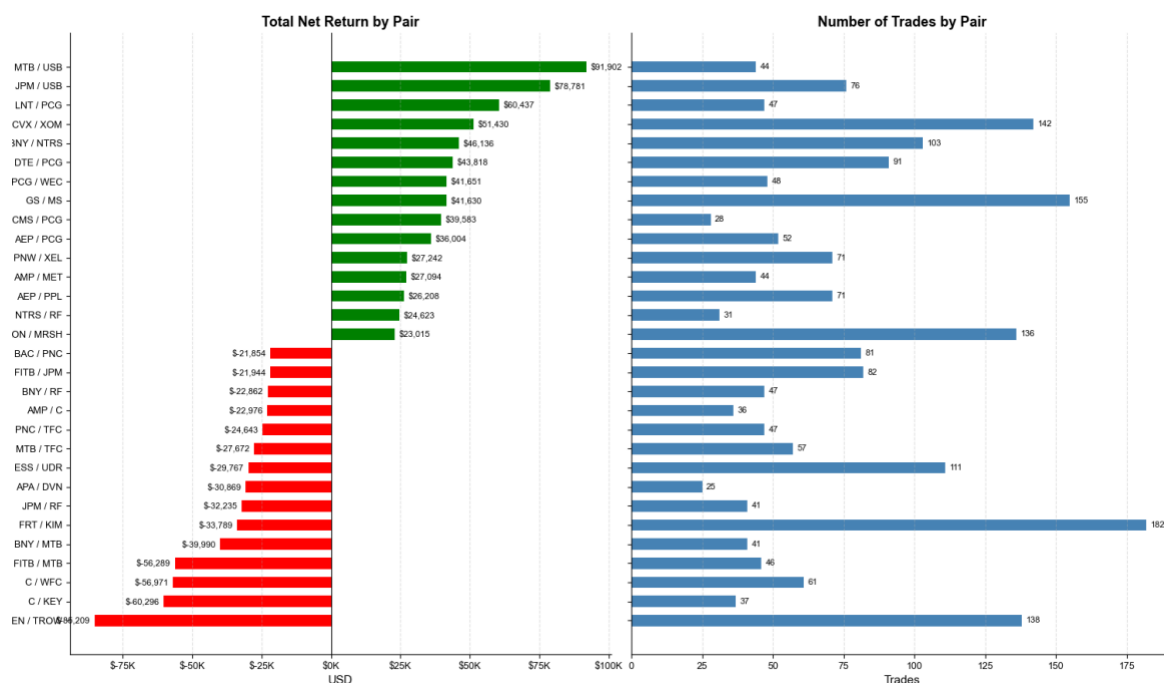
3 Results

Statistics / Test Periods	2015 - 2018	2018 - 2021	2021 - 2024
Net P&L (USD)	-84,779	626,921	896,732
Max Drawdown (USD)	-505,115	-272,114	-990,206
Return / Max Drawdown	-0.17	2.30	1.00
Sharpe Ratio	-0.15	0.98	0.44
Trades	13,033	628	2,146
Win Rate	56.49%	59.90%	54.75%

Initial Capital: 1,000,000

3.1 The 2015 – 2018 Test Period

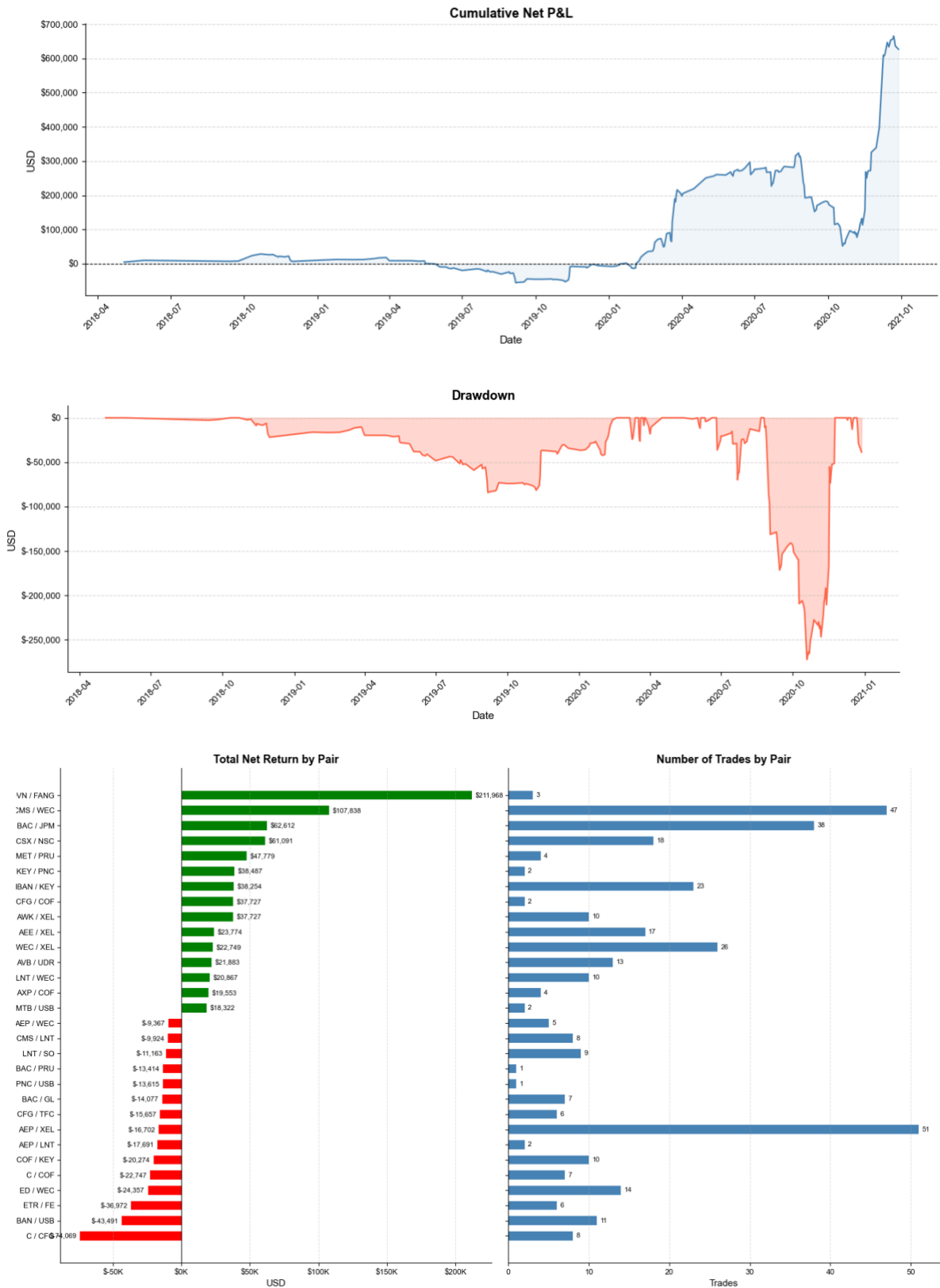




2015 – 2018 test period can be viewed as the overfitting case. We have a total net loss of \$85,000, a negative Sharpe, and 13,033 trades. That is, nearly 17 trades daily, and 20 times more trades compared to the 2018 – 2021 test period. The net cumulative P&L chart peaked near \$200,000 in mid-2016, then went through a series of drawdowns, bottoming near -\$300,000 in late 2017.

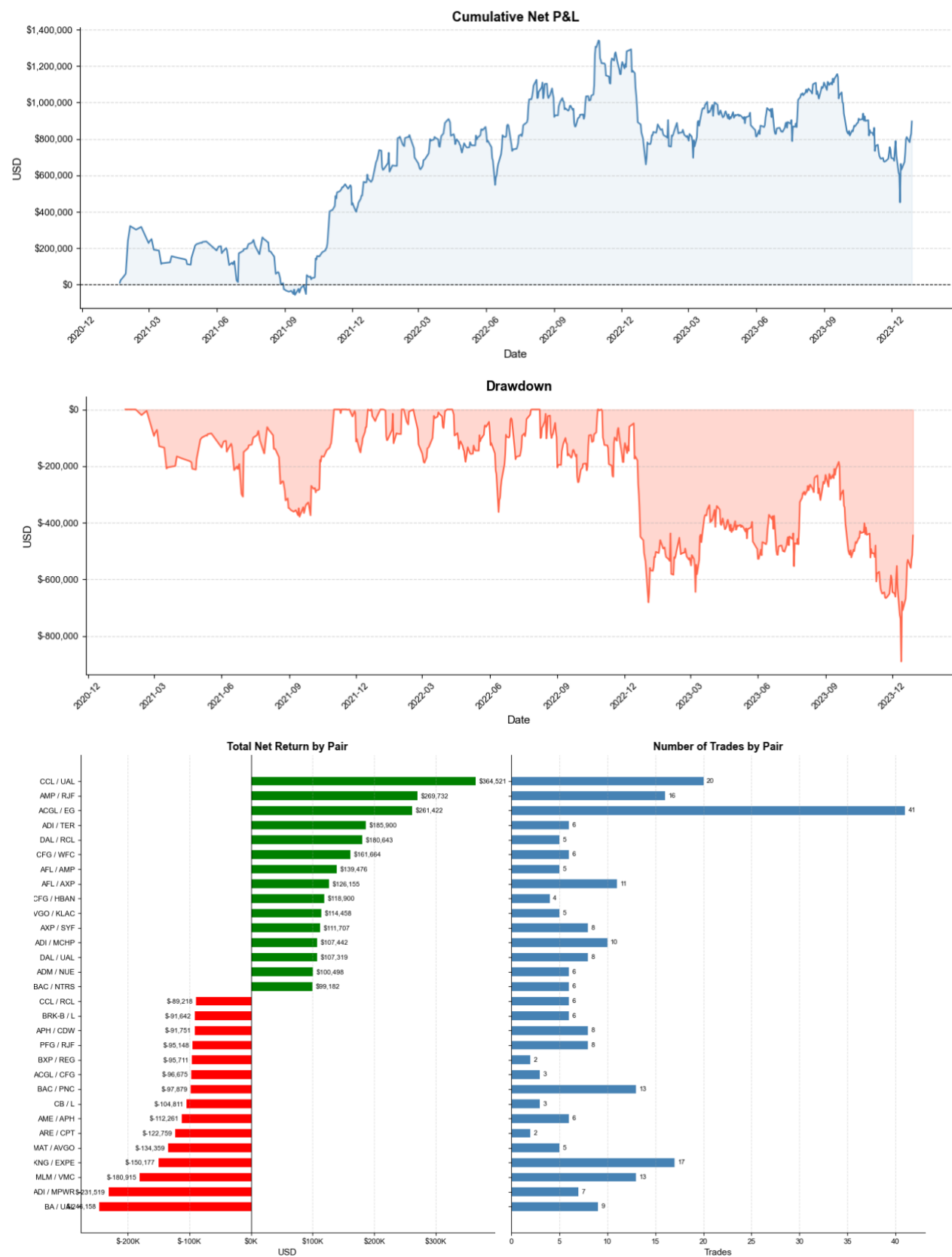
Here, high-churn parameters selected a large number of cointegrated pairs while setting a low entry z-score threshold, leading to high trade counts, and the strategy bled out in slippage. The exit type evidences this claim; this period had nearly 900 burst (stop-loss) exits versus ~27 in the other two, meaning far more positions diverging rather than mean-reverting. High frequency trade, high stop-out rate, and a negative net, all indicating over-optimized parameter configuration and strategy.

3.2 The 2018 – 2021 Test Period



This is the strongest regime out of the three. Highest return/max drawdown ratio of 2.30, best Sharpe of 0.98, highest win rate of 59.9%, all achieved with the fewest number of trades of 628. Yet, most of this period's profit is concentrated in the COVID-volatility window of 2020, and the pair chart visualizes that few pairs, including the DVN/FANG pair with 3 trades, dominate most profits. In other words, although this strategy outputs the best performance metrics, its success is concentrated in a few pairs in a high-dispersion regime, not a steady alpha. That said, it is noteworthy that this strategy did not suffer the deep drawdowns seen in the other two periods.

3.3 The 2021 – 2024 Test Period



This period gained profit with severe drawdown. It scored highest dollar P&L of \$896,732, but a maximum drawdown of -\$990,206, and a middling risk-adjusted quality of 1.00 return / drawdown and a 0.44 Sharpe. The chart climbs up to ~\$1,400,000 by late 2022, then bleeds down through 2023. It is notable that the BA/UAL pair lost \$246,158 while CCL/UAL made \$364,521, i.e., the same ticker UAL gained the most net return while losing the most.

This strategy appears to be a middling strategy among the three test periods. It had a medium number of trades and gained the most profit, but at the cost of the deepest drawdown of the three. It must be considered that this was a volatile regime, which includes the post-pandemic and the 2022 rate-hiking cycle, and it is questionable whether this configuration could stay profitable in a calmer regime, the kind the 2018-2021 setup weathered without large losses.

3.4 Patterns and Findings

Trade frequency appears to track performance inversely. 628 trades produced the best results, 2,146 a mediocre outcome, and 13,033 a loss. With only three data points, each from a different regime and different optimized parameter set, this is a suggestive pattern rather than a causal one. Higher trade counts may simply coincide with the unfavorable regime. Still, the consistency of the direction across all three periods is suggestive.

Returns are regime-dependent. The strategy performs when cross-sectional dispersion and market volatility are high (2020 COVID, 2022 rate shock), and stalls or loses in calmer markets. Within each period, especially in volatile regimes, a few pairs drive the most P&L.

Drawdowns are severe relative to returns in two of three periods. The optimizer used a simplified Sharpe that penalizes volatility symmetrically rather than capturing asymmetric tail losses, so it did not select against these deep drawdowns.

Win rates (54%~60%) are consistent with a weak mean-reversion edge. A win rate slightly above a coin flip, combined with the deep 2021-2024 drawdown, indicates that the losing trades are larger than the winning ones: an asymmetric payoff typical of mean-reversion.

Taken together, the three periods suggest a genuine but marginal mean-reversion edge that depends heavily on volatile, high-dispersion regimes and carries drawdown risk the optimizer did not control for.

4. Limitations

4.1 Simplified Sharpe

The metric used to optimize the parameters was the simplified Sharpe, built from zero-filled exit-date P&L, not a mark-to-market equity curve. Because the optimizer maximized this metric, its inaccuracy directly shaped the selected parameters. Hence, implementing a proper daily mark-to-market equity curve, and combining it with other metrics including the Sortino ratio, would provide a more reliable optimization objective.

4.2 No Drawdown Control

The optimizer used a volatility-symmetric Sharpe that did not penalize the deep drawdowns. Furthermore, there was no portfolio-level stop, no volatility targeting, no exposure cap beyond the per-trade budget. This was significant as two of three periods had drawdowns near or exceeding their total gains. This would be fixed by optimizing the parameters using a drawdown-aware objective instead of the simplified Sharpe ratio, and adding a portfolio-level drawdown management strategy.

4.3 Transaction Costs

The strategy has applied a fixed 0.05% per transaction leg. A flat cost was used as a simplification; real-world costs vary with order size, liquidity, and volatility. More realistic implementation of transaction costs would make the backtest more realistic, as the 2015-2018 period showed, where ~13,000 trades bled the strategy into a net loss largely through costs.

4.4 Survivorship Bias

The strategy crawls data from S&P 500 constituents from Wikipedia and yfinance. However, the index composition today differs from its 2015 composition, and the strategy naturally dropped tickers that were later delisted or removed. This is compounded by dropping any ticker with missing data over the period, which further restricts the universe to names with complete, surviving histories. In other words, the strategy has only made trades from tickers that survived to today. This could be fixed by implementing a point-in-time index membership.

4.5 Cointegration Bias

The cointegration test checks both directions, Y on X and X on Y . This adds a mild bias in which some pairs pass that a single fixed-direction test would have rejected. The logic tests both directions and keeps whichever yields the lower Augmented Dickey-Fuller p-value, to catch cointegration regardless of which asset is treated as dependent. This could be addressed by requiring a pair to pass in both directions, or by using the Johansen test, which is direction-symmetric by construction.

4.6 Conclusion

Across three out-of-sample periods, the strategy was profitable in two but showed no robust, regime-dependent edge. Its returns concentrated in volatile, high-dispersion markets and a small number of pairs, and it carried drawdowns the optimizer did not control for. Whether the marginal edge observed here could become durable depends largely on the limitations above, particularly realistic transaction costs and proper drawdown-aware risk management.

The project also yielded broader lessons. First, the centrality of risk management: avoiding deep drawdowns matters as much as capturing profitable trades, and the periods that managed risk best were not always those that earned the most. Second, the weight of the optimization design itself. The choice of objective metric and which parameters to optimize shaped outcomes as much as the trading logic did. A flawed Sharpe propagated directly into the selected parameters. Finally, this was my first application of statistical tools from coursework to a real-world dataset, and their effectiveness was impressive. Filtering cointegrated pairs from the full S&P 500 in seconds demonstrated how much these methods can do when applied correctly.